

# Are You Getting What You Pay For?

## Auditing Model Substitution in LLM APIs

Software-based detection is fundamentally unreliable. Hardware attestation is the only viable path to verifiable inference.

[arXiv:2504.04715](https://arxiv.org/abs/2504.04715) • [ML Research](#) / [AI Security](#) / [Cloud Infrastructure](#)

# The Model Substitution Problem: Economic Incentives Drive Covert Swaps

## Core Threat

- API providers charge for specific models (e.g., Llama-3-70B, Qwen2-72B).
- Black-box APIs give users no guarantee the advertised model is actually served.
- Strong economic incentives exist to swap in cheaper alternatives—quantized, smaller, or different architectures—while maintaining pricing.

## Substitution Types & Formal Framework

1. Smaller models (e.g., 8B instead of 70B)
2. Quantized variants (FP8/INT8 instead of FP16)
3. Fine-tuned or updated versions
4. Entirely different model families

Formalized as hypothesis testing:

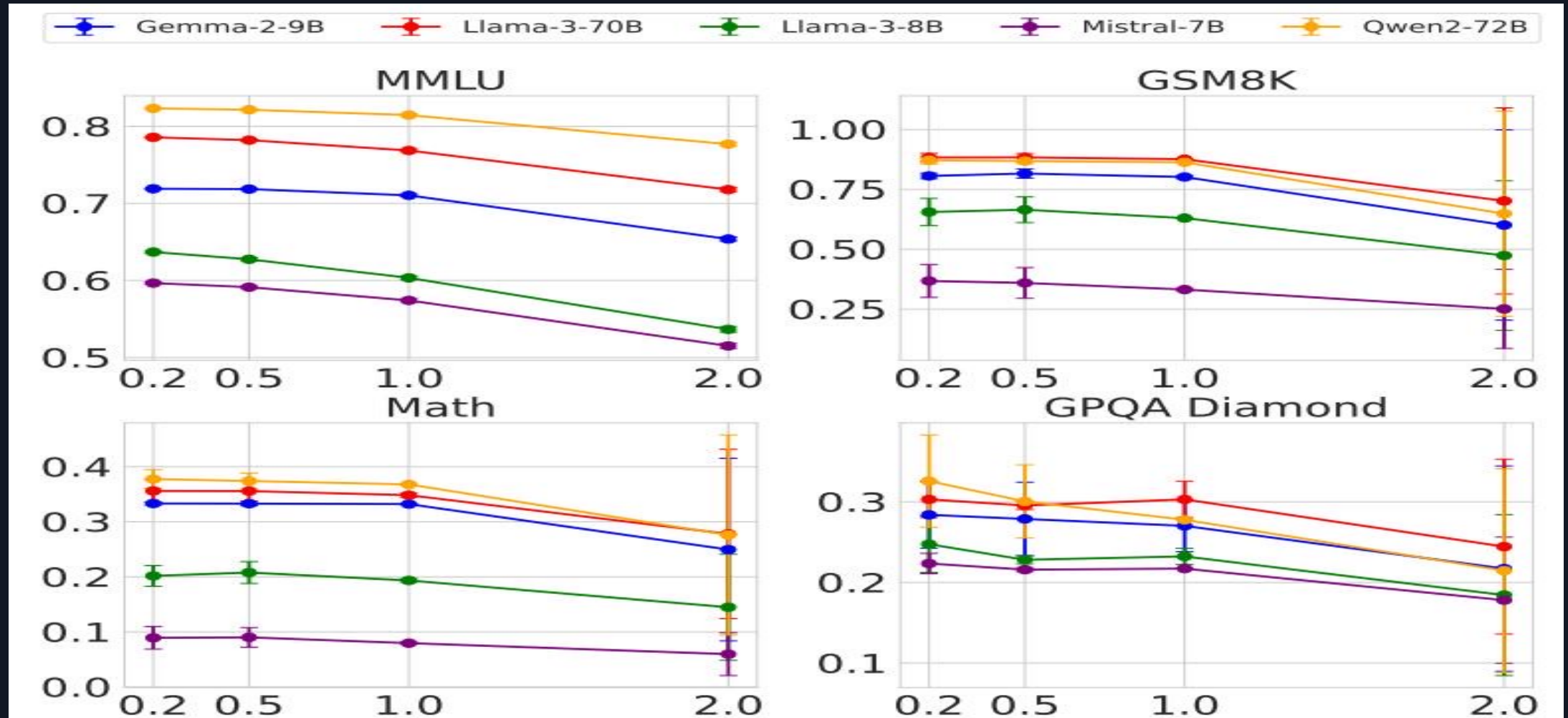
$H_0$ : API outputs  $\sim P_{\text{spec}}(y|x)$  vs  $H_1$ : API outputs  $\sim P_{\text{actual}}(y|x)$

# FP8 Quantization Preserves Most Accuracy — Making Substitution Hard to Detect

| Model                        | MMLU         | GSM8K        | MATH         | GPQA Diamond |
|------------------------------|--------------|--------------|--------------|--------------|
| Llama-3-8B-Instruct          | 62.69 ± 0.18 | 61.14 ± 3.47 | 20.65 ± 3.43 | 22.62 ± 0.22 |
| Llama-3-8B-Instruct-FP8      | 62.43 ± 0.26 | 60.90 ± 4.10 | 14.91 ± 2.49 | 20.14 ± 0.24 |
| Llama-3-70B-Instruct         | 78.05 ± 0.08 | 88.06 ± 1.44 | 35.69 ± 1.33 | 29.60 ± 0.51 |
| Llama-3-70B-Instruct-FP8     | 77.88 ± 0.13 | 87.35 ± 1.24 | 35.75 ± 1.16 | 33.12 ± 0.30 |
| Gemma-2-9b-it                | 71.86 ± 0.08 | 81.80 ± 1.35 | 33.41 ± 0.28 | 28.64 ± 2.97 |
| Gemma-2-9b-it-FP8            | 71.92 ± 0.11 | 79.41 ± 1.14 | 32.53 ± 0.34 | 27.81 ± 3.22 |
| Qwen2-72B-Instruct           | 82.18 ± 0.08 | 86.72 ± 1.00 | 37.39 ± 1.41 | 29.93 ± 2.71 |
| Qwen2-72B-Instruct-FP8       | 81.98 ± 0.08 | 86.82 ± 0.97 | 37.67 ± 1.39 | 31.08 ± 1.96 |
| Mistral-7B-Instruct-v0.3     | 59.15 ± 0.10 | 35.90 ± 4.54 | 8.94 ± 1.24  | 21.60 ± 0.17 |
| Mistral-7B-Instruct-v0.3-FP8 | 58.77 ± 0.13 | 32.20 ± 4.02 | 7.68 ± 1.23  | 22.72 ± 0.19 |

Observation: For most models, FP8 scores fall within the error bars of the original, especially on MMLU and GSM8K.

# Higher Temperature Amplifies Performance Gaps but Increases Noise



At temperature 2.0, performance degrades across all models. The gap between original and substitute narrows or becomes noisy, complicating statistical detection.

# Log Probability Fingerprinting Fails Due to Inference Nondeterminism

| Model                       | BERT Acc | T5 Acc | GPT2 Acc | LLM2Vec Acc |
|-----------------------------|----------|--------|----------|-------------|
| Llama3-70B-Instruct-FP8     | 50.55    | 50.10  | 50.45    | 49.90       |
| Llama3-70B-Instruct-INT8    | 51.60    | 49.90  | 51.30    | 50.25       |
| Gemma2-9b-it-FP8            | 49.95    | 50.30  | 51.20    | 49.80       |
| Gemma2-9b-it-INT8           | 49.00    | 49.70  | 51.65    | 49.55       |
| Mistral-7b-v3-Instruct-FP8  | 50.55    | 49.75  | 48.70    | 48.75       |
| Mistral-7b-v3-Instruct-INT8 | 49.70    | 50.75  | 50.50    | 51.15       |
| Qwen2-72B-Instruct-FP8      | 50.05    | 50.25  | 48.75    | 49.80       |
| Qwen2-72B-Instruct-INT8     | 50.75    | 50.55  | 49.45    | 50.20       |

The same Llama-3-8B model produces different log probabilities across software versions (transformers 4.40 vs 4.50, vLLM 0.6.2 vs 0.8.2) and hardware (A100 vs H100).

# Randomized Routing at Low Rates Makes Statistical Detection Query-Prohibitive

## Attack Model

- Provider routes only a fraction  $p$  of queries to the cheaper substitute.
- Mixed output distribution:  $(1-p) \cdot P_{\text{spec}} + p \cdot P_{\text{sub}}$
- As  $p \rightarrow 0$ , the mixed distribution converges to  $P_{\text{spec}}$ .
- Auditor must distinguish an ever-smaller signal from noise.
- Query budget grows super-linearly (often exponentially) as  $p$  decreases.

## Implication

- Low-rate substitution is economically attractive (small savings per query, massive aggregate gain).
- Statistical tests (e.g., MMD) require impractically large sample sizes.
- Benchmark evasion: detect known evaluation prompts and route them to the genuine model.
- Result: software-only auditing is query-prohibitive and easily gamed.

# Software-Only Auditing Methods Are Fundamentally Unreliable

| Method Type                              | Weakness                                                                                                          | Practical Limitation                                                                                          |
|------------------------------------------|-------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------|
| Text-based statistical tests (e.g., MMD) | Require impractically large query budgets; fail against subtle or randomized substitutions.                       | Query-prohibitive at low substitution rates; cannot distinguish within error-bar performance gaps.            |
| Log-probability fingerprinting           | Production inference environments introduce significant nondeterminism across software versions and GPU hardware. | Same model on different infrastructure produces divergent log-prob signatures; false positives dominate.      |
| Adversarial countermeasures              | Providers can actively evade: randomized routing + benchmark evasion.                                             | Benchmark prompts routed to genuine model; regular traffic substituted. Mixed distribution approaches target. |

# Trusted Execution Environments Provide Provable Cryptographic Model Integrity

## What TEEs Guarantee

- Hardware-secured enclaves with cryptographic attestation of code and data.
- Provider proves specific model weights are loaded and correct inference code is running.
- Attestation is tamper-evident and verifiable by the client.
- Only modest performance overhead vs. standard inference.

## Why TEEs Win vs. Software-Only

- Provable: cryptographic proof, not statistical inference.
- Efficient: modest overhead, practical for large models today.
- Robust: immune to randomized routing and benchmark evasion.
- Deployable: NVIDIA Confidential Computing on H100 GPUs is production-ready.
- Contrast with ZKPs: computationally prohibitive for LLM-scale inference.



# Key Takeaways: Close the Verification Gap with Hardware Attestation

1. Model substitution is economically motivated and practically undetectable via software-only means.
2. FP8 quantization barely affects benchmark scores (often within error bars), making it a nearly invisible substitute.
3. Inference nondeterminism across software versions and GPU hardware defeats log-probability fingerprinting.
4. Randomized substitution at low rates converges to the target distribution, rendering statistical tests query-prohibitive.
5. Trusted Execution Environments are the only currently viable path to verifiable LLM APIs — provable, efficient, and deployable today.

**Call to action: Integrate TEE-based attestation into API trust frameworks and procurement requirements.**